

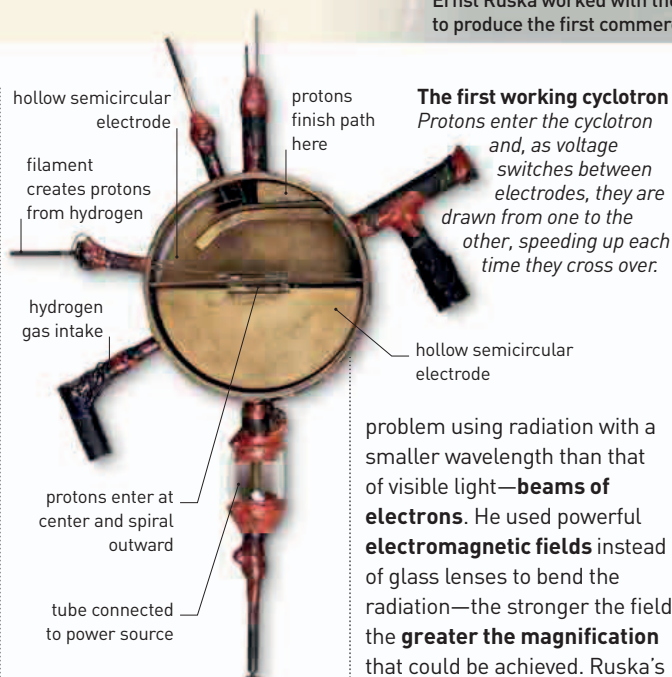
“IT IS A **MIRACLE** THAT... **DIFFICULTIES** HAVE BEEN SOLVED TO AN EXTENT THAT SO **MANY SCIENTIFIC DISCIPLINES...** CAN **REAP ITS BENEFITS.**”

Ernst Ruska, Nobel lecture on electron microscopy, December 8, 1986



Ernst Ruska worked with the German company Siemens and Halske to produce the first commercial electron microscope in 1939.

THE 1930S WITNESSED THE BIRTH OF TECHNOLOGIES that would help reveal the secrets of the microscopic world. Physicists had devised ways of firing charged particles at very high speeds so that they could study resulting collision products. They did this with accelerators that shot particle beams through evacuated tubes and used electric fields to keep them going. In 1931, American physicist **Ernest Lawrence** (1901–58) invented a new kind of **particle accelerator** with a spiral center called a **cyclotron**, designed to shoot hydrogen ions. His first model had a diameter of 5 in (12.5 cm) and energized ions with **80,000 electron volts** (energy acquired by an electron as it accelerates through the potential of one volt). Lawrence's team went on to make ever more powerful cyclotrons. By 1946, his laboratory at Berkeley, California,



The first working cyclotron
Protons enter the cyclotron and, as voltage switches between electrodes, they are drawn from one to the other, speeding up each time they cross over.

problem using radiation with a smaller wavelength than that of visible light—**beams of electrons**. He used powerful **electromagnetic fields** instead of glass lenses to bend the radiation—the stronger the field, the **greater the magnification** that could be achieved. Ruska's **electron microscope** allowed extremely large magnifications, and scientists could examine tiny molecules, possibly even atoms.

had built a 173-in- (440 cm-) device powered by **100 million electron volts**.

The particles targeted by physicists are **billions of times smaller than the smallest objects that can be seen** with light microscopes. Until 1930, all microscopes magnified images using conventional optics that refract (bend) light rays using lenses. However, the wavelength of light, which is measured in 10 thousandths of a millimeter, restricts the amount of detail that can be seen in objects that approach this size. German physicist **Ernst Ruska** (1906–88) found a radical way to solve this

“IT MAY BRING TO LIGHT... A **DEEPER** KNOWLEDGE OF THE **STRUCTURE** OF MATTER.”

Ernest Lawrence, Nobel Prize speech, November 29, 1940



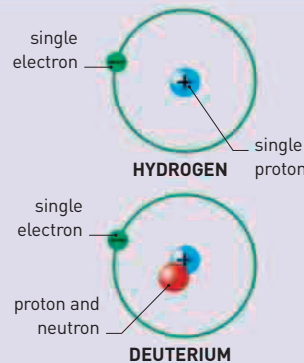
Between these bright galaxies lies invisible dark matter.

ERNEST RUTHERFORD HAD PREDICTED the existence of a second proton-sized subatomic particle (see 1919). In 1932, British physicist **James Chadwick** (1891–1974) examined a type of radiation that could knock protons from atoms and found Rutherford's **missing particle: the neutron**. Atomic capabilities advanced in April, when British and Irish physicists **John Cockcroft** (1897–1967) and **Ernest Walton** (1903–95) split lithium atoms **into helium atoms** (see panel, opposite).

British physicist Paul Dirac had predicted the existence of anti-particles of electrons (see 1928). Confirmation came in August 1932 when American physicist **Carl Anderson** (1905–91) studied trails of charged particles in a detector called a cloud chamber and found that some electron-like particles were **positive instead of negatively charged**.

DEUTERIUM, OR HEAVY HYDROGEN

Heavy hydrogen (deuterium) accounts for fewer than one in 6,000 of Earth's hydrogen atoms. All hydrogen atoms have a single proton (positively charged particle), so have an atomic number of 1 (see panel, 1913). Normal hydrogen has only one particle in its nucleus. Deuterium has an additional particle—a neutron (particle with no charge)—in its nucleus.



600,000
THE NUMBER
OF VOLTS
NEEDED TO
SPLIT THE
ATOM

January 2
Ernest Lawrence
uses the first
cyclotron

March 3 Ernest
Ruska and German
engineer **Max Knoll**
achieve the first electron
optical magnification with
the prototype of the
electron microscope

August 1
American botanists
Harriet Creighton and
Barbara McClintock
explain **genetic**
recombination by shuffling
of chromosome fragments

French physicist **Irène**
Joliot-Curie demonstrates
radiation that she thinks is
gamma rays, but **James Chadwick**
later shows it to be **neutrons**

January 1,
1932 German
physicist **Werner**
Heisenberg begins the
proton-neutron model of
the nucleus and uses it to
explain isotopes

August American
physicist **Karl Guthe**
Jansky discovers
radio waves coming
from the **Milky**
Way galaxy

November 26 Harold
Clayton Urey discovers
heavy hydrogen, later
called **deuterium**

January 1932
Dutch-American biologist
Cornelius van Niel suggests
that plants use water as an
electron source for
photosynthesis